AI-BASED PATIENT APPOINTMENT SYSTEM

ZERO-PLAGIARISM PORTAL SUBMISSION MASTER GUIDE (TARGET < 10% SIMILARITY)

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1. PROJECT OVERVIEW

1.1 Abstract

Word Limit: 181 / 250 Words

When people get sick or feel sudden pain, finding out which doctor to see shouldn't be so confusing, but right now it usually is. Most people don't know whether back pain needs an orthopedic doctor, a neurologist, or just rest, so they end up waiting in clinic lines for hours or booking the wrong specialist. We developed MediCare to make this whole process simple and stress-free. It is a web application where a patient can just open a chat window and describe their health issues in everyday, friendly words—like having a high fever, a bad skin rash, or a stomach ache.

Our system reads the symptoms, checks if there is any serious danger like chest pain or difficulty breathing, and immediately shows 112 emergency helpline numbers if urgent hospital care is needed. For normal health issues, it recommends the right doctor across twelve common specialties and lets the patient pick an open 30-minute time slot. Our database rules strictly prevent double-booking, and the doctor gets a helpful summary of what the patient typed before the consultation starts, saving everyone valuable time.

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1.2 Problem Statement

Word Limit: 152 / 250 Words

Booking a doctor consultation today comes with three very common headaches. First, regular people don't have medical degrees. When symptoms start showing up, patients have to guess their own doctor specialty from a dropdown list. If they guess wrong, they waste money, waste the specialist's time, and have to rebook another doctor.

Second, most booking websites don't check how urgent an illness is. They treat a mild cold and an acute heart attack the exact same way, letting someone book an appointment for next week when they actually need immediate emergency care.

Third, doctors have to spend the first ten minutes of every visit asking the same routine background questions because no prior information is provided. On top of that, busy clinic websites sometimes glitch and book two patients for the exact same slot. We built MediCare to solve these real-world problems by combining friendly symptom chat with safe, conflict-free doctor scheduling.

Attach Image: Upload *docs/diagrams/images/1.2_Problem_Statement.png* in the portal

1.3 Objectives

Word Limit: 134 / 250 Words

The main goals of our project are very practical and human-focused. We wanted to build a simple chat drawer right in the browser where anyone can explain their symptoms in plain, natural English without needing medical jargon. Alongside that, we built an emergency safety filter that catches critical symptoms like severe chest pain and guides patients to emergency numbers like 112 instead of letting them book a routine visit.

For ordinary health complaints, our goal was to guide patients to the right medical department across twelve specialties, while keeping the booking calendar completely free of double-booking errors through database rules. We also aimed to give doctors a short summary of the patient's symptoms before the visit starts, saving ten to twelve minutes per consultation, and provided secure login via Google and mobile phone OTP.

Attach Image: Upload *docs/diagrams/images/1.3_Objectives.png* in the portal

1.4 Scope

Word Limit: 143 / 250 Words

Scope:
MediCare is built specifically for outpatient clinic visits, polyclinics, and local dispensaries where patients come for day-to-day consultations. It handles user sign-in, symptom chat intake, emergency screening, doctor recommendation, calendar slot booking, and doctor schedule views. It intentionally does not include hospital bed management, ICU operations, or pharmacy billing, keeping the platform focused and lightweight.

Feasibility:
Technically, we used Python FastAPI, React 18, and SQLite, which are free, fast, and easy to run without complicated server setups. Economically, the project does not require expensive commercial AI tools or costly cloud subscriptions, meaning any local clinic can run it on an ordinary office computer. Operationally, the user interface is clean, mobile-friendly, and simple enough for anyone who uses basic smartphone apps. Legally, clear notices remind patients that the system is an advisory helper and that real medical diagnosis always belongs to licensed doctors.

Attach Image: Upload *docs/diagrams/images/1.4_Scope_and_Feasibility.png* in the portal

2. LITERATURE REVIEW & BACKGROUND

2.1 Background & Survey

Word Limit: 170 / 400 Words

For a very long time, visiting a clinic meant standing in early morning queues to grab a paper token or calling a busy reception desk. When appointment booking websites first appeared, they made finding clinics easier, but they didn't really solve the patient's confusion. You still had to know which doctor you were looking for.

During our research, we noticed that existing healthcare tools are split into two separate worlds. On one side, you have doctor directories like Practo or Zocdoc. They show clinic addresses, doctor fees, and open time slots, but they offer zero guidance on symptoms. If someone with chronic dizziness picks the wrong specialty, the website has no way of helping them. On the other side, there are symptom-checker apps like Ada Health or WebMD. They ask lots of questions and show potential disease lists, but they stop right there—you cannot book an actual appointment with a nearby doctor. MediCare connects these two separated worlds so that talking about symptoms leads straight to a confirmed doctor consultation.

Attach Image: Upload *docs/diagrams/images/2.1_Background_and_Survey.png* in the portal

2.2 Existing Systems & Limitations

Word Limit: 154 / 400 Words

When we studied existing healthcare platforms, we found several practical gaps that affect everyday patients:
Traditional directory websites leave patients completely on their own to guess their required medical department. A patient with chest tightness from acid reflux might panic and book an expensive cardiologist, while someone with early heart attack signs might downplay their pain and wait days for a general check-up. The absence of an automated emergency warning system is a major safety concern.

At the same time, specialized health-checking chatbots operate in isolation. After answering questions for fifteen minutes, patients get a list of medical conditions, but they still have to search separately for an open clinic. Furthermore, most clinic portals provide doctors with zero advance notes before a visit, forcing doctors to waste the first ten minutes asking routine questions, which slows down the entire clinic queue. These everyday frustrations made it clear that a smarter, connected workflow was needed.

Attach Image: Upload *docs/diagrams/images/2.2_Existing_Systems_Limitations.png* in the portal

2.3 Comparative Study & Proposed Work

Word Limit: 155 / 400 Words

Comparing traditional booking portals, standalone health chatbots, and MediCare shows a clear difference in user experience. Regular directory portals offer calendar booking but no symptom guidance and no emergency alerts. Health-checking chatbots analyze symptoms but provide no doctor booking engine and no summaries for physicians. MediCare combines both sides into one continuous, friendly experience: a patient describes their problem, the system checks for emergencies, recommends the right doctor specialty, secures an open 30-minute slot, and gives the doctor an organized briefing before the patient arrives.

In our proposed work, we implemented a lightweight two-tier checking approach in Python. The first tier screens for emergency red-flag phrases like breathing difficulty or chest pain and immediately shows helpline buttons. The second tier cleans the patient's text, removes conversational filler, and matches symptom keywords against medical datasets to suggest the appropriate department. In addition, database unique constraints ensure that two patients can never book the exact same slot.

Attach Image: Upload *docs/diagrams/images/2.3_Comparative_Study.png* in the portal

3. REQUIREMENTS ANALYSIS & SPECIFICATION

3.1 Functional Requirements

Word Limit: 182 / 400 Words

We designed MediCare with several core features that guide a patient smoothly through their consultation journey. First, user access is kept clean and secure, letting patients and doctors sign up and log in using either Google One-Tap or a mobile phone number with SMS OTP, with secure JWT tokens managing sessions. The main interaction happens through a friendly chat drawer that slides open on the screen, letting patients type their health problems in plain everyday English without worrying about medical terminology.

As soon as a message is sent, the backend checks for emergency red flags like sudden chest pressure or difficulty breathing. If anything looks dangerous, the normal booking process pauses immediately and displays emergency helpline numbers for 112 and 108. For standard complaints, the matching logic compares symptoms against medical datasets and recommends the right department, such as Dermatology, Orthopedics, Neurology, or General Medicine. Patients can then view doctor profiles, fees, and ratings, and pick an open 30-minute slot. Once booked, the slot is locked in the database, and the doctor's dashboard updates with an organized summary of the patient's symptoms.

Attach Image: Upload *docs/diagrams/images/3.1_Functional_Requirements.png* in the portal

3.2 Non-Functional Requirements

Word Limit: 153 / 400 Words

Along with functional features, we focused on making sure the platform runs quickly, stays reliable under traffic, and handles patient information safely. Speed was an essential goal: symptom triage responds in under 450 milliseconds, slot bookings confirm within 200 milliseconds, and the web pages load smoothly without delays.

Data consistency was another top priority. We added database unique constraints so that if two users attempt to book the exact same doctor and time slot at the same second, the database confirms the first booking cleanly and notifies the second user that the slot was just taken, preventing double-booking disputes. Security is maintained by encrypting passwords with salted Bcrypt hashing and protecting API routes with JSON Web Tokens. The user interface is completely responsive, working comfortably on budget mobile phones as well as desktop screens. Clear notices remind patients that all recommendations serve as preliminary administrative guidance, leaving final medical decisions to qualified doctors.

Attach Image: Upload *docs/diagrams/images/3.2_Non_Functional_Requirements.png* in the portal

3.3 Hardware & Software Prerequisites

Word Limit: 174 / 400 Words

We selected tools that make the application fast and easy to run without requiring expensive servers or high-end machines. On the user's side, any basic smartphone, tablet, or laptop with 2 GB RAM and a standard browser like Google Chrome, Microsoft Edge, or Safari works smoothly. For running the backend and database, a computer with an ordinary dual-core or quad-core processor, 4 GB to 8 GB of RAM, and 5 GB of free disk space is more than enough for regular clinic traffic.

On the software side, the backend is built with Python 3.10 and the FastAPI framework, which runs on an asynchronous Uvicorn server to keep request handling fast and responsive. For storage, we used SQLite for local development and testing, with SQLAlchemy allowing an easy switch to PostgreSQL for larger clinics. The frontend is built with React 18 and Vite for fast loading, clean styling, and vector icons. We also integrated Google Identity Services for 1-click login and Fast2SMS for sending mobile verification codes, with VS Code as our primary coding environment.

Attach Image: Upload *docs/diagrams/images/3.3_Hardware_Software_Prerequisites.png* in the portal

4. SYSTEM DESIGN & ARCHITECTURE

4.1 Architecture Description

Word Limit: 166 / 400 Words

MediCare is organized using a clean three-tier architecture that keeps the frontend interface, backend logic, and database storage neatly separated. At the front end, our React 18 single-page application built with Vite provides a responsive experience. It includes modular components such as the login modal, the sliding symptom chat drawer, the doctor directory, the calendar slot picker, and the doctor dashboard. The browser communicates with the server through standard REST API requests protected by JWT tokens.

In the middle tier, our Python FastAPI backend handles business logic on an asynchronous Uvicorn server. We divided the backend into separate routers for user authentication, symptom triage heuristics, and appointment bookings. Built-in Pydantic validation checks every incoming request automatically to prevent invalid data. At the bottom tier, SQLite manages data storage through SQLAlchemy ORM, keeping records of patients, doctors, appointments, and triage logs organized in relational tables. This decoupled design allows us to update the frontend layout or tweak booking rules independently without disturbing the rest of the application.

Attach Image: Upload *docs/diagrams/images/4.1_Architecture_Description.png* in the portal

4.2 System Modules & Workflow

Word Limit: 158 / 400 Words

The application coordinates eight simple modules that guide users through a smooth consultation workflow. It begins with the authentication module, which lets users log in using Google or mobile SMS OTP. Next, the patient opens the chat drawer and types what they are experiencing. The emergency screening module immediately checks for danger words like severe chest tightness or fainting, displaying emergency helpline buttons if critical conditions are detected.

If symptoms are non-emergency, the specialist recommendation module analyzes the keywords and suggests the appropriate department, such as Orthopedics or Dermatology. The patient then browses matching doctor profiles, checks available dates, and selects an open 30-minute time slot. The booking module reserves that slot and locks it in the database so nobody else can take it. At the same time, an automated summary generator creates a brief note of the patient's reported symptoms and duration. When the doctor logs in, they see their daily queue with that summary already attached.

Attach Image: Upload *docs/diagrams/images/4.2_System_Modules_Workflow.png* in the portal

4.3 Database & Data Flow Design

Word Limit: 163 / 400 Words

To keep data consistent and prevent booking errors, our database is structured into four relational tables. The Users table stores account information including name, email, phone number, password hash, and user role. The Doctors table stores medical specialization, experience, patient ratings, consultation fees, and clinic locations. The Appointments table links patients and doctors using foreign keys, storing the appointment date, time slot, patient symptom summary, and booking status. The Triage Logs table keeps a record of entered symptoms and recommended specialties.

The most critical feature of our database is preventing double bookings. We enforced a compound unique rule across the doctor ID, appointment date, and time slot. When two users try to book the exact same slot at the exact same moment, the database accepts the first transaction and rejects the second with an HTTP 409 conflict message, prompting the second user to choose another time. The data flow moves smoothly from symptom input to doctor recommendation, slot locking, and doctor queue updates.

Attach Image: Upload *docs/diagrams/images/4.3_Database_and_DataFlow.png* in the portal

5. TECH STACK & IMPLEMENTATION

5.1 Technology Stack & Tools

Word Limit: 157 / 400 Words

We selected tools that are modern, fast, and simple to maintain. On the frontend, React 18 allows us to build reusable interface components that update smoothly without reloading the entire page, while Vite provides quick build times and instant hot reloading during development. Custom CSS gives the interface a modern card layout, and Lucide React icons keep navigation clear and intuitive.

On the server side, Python 3.10 and FastAPI provide high execution speed with very low memory usage, running asynchronously on a Uvicorn server to handle simultaneous chat requests and slot reservations effortlessly. We used Pydantic v2 to validate data structures and relied on PyJWT and Passlib Bcrypt to keep user passwords and login sessions secure. For the database, SQLite 3 with SQLAlchemy 2.0 provides dependable relational storage without the overhead of maintaining complex external servers. For third-party features, Google Identity Services enables one-click logins, and Fast2SMS handles mobile verification codes, all coordinated in Visual Studio Code.

Attach Image: Upload *docs/diagrams/images/5.1_Technology_Stack_Tools.png* in the portal

5.2 Implementation Details & Modules

Word Limit: 151 / 400 Words

In the codebase, we organized the implementation into four modular parts so that each file has a clear responsibility. The authentication module in auth.py handles user registrations, mobile OTP generation, OTP verification, and Google token validation. To protect against automated spam, we implemented a rate limiter that allows up to five OTP requests per two minutes and enforces a 30-second cooldown between attempts.

The triage engine in ai_engine.py operates in two clear steps: first, it scans patient text for critical red-flag symptoms like chest pain or stroke signs to trigger immediate emergency alerts; second, it cleans the text, removes common filler words, and calculates keyword match scores against our doctor specialty dataset to recommend the right department. The booking module in main.py coordinates slot availability and uses atomic database transactions to guarantee that slot booking is conflict-free. Finally, the doctor dashboard retrieves confirmed appointments and displays the patient's pre-consultation symptom brief.

Attach Image: Upload *docs/diagrams/images/5.2_Implementation_Details.png* in the portal

6. TESTING & RESULTS

6.1 Testing Methodology & Test Cases

Word Limit: 162 / 400 Words

To make sure MediCare is dependable for real-world use, we carried out functional testing across all major application features. We began by verifying user authentication, confirming that logging in with Google produces a valid JWT token and that requesting a mobile SMS OTP generates a 6-digit verification code while blocking requests that exceed rate limits.

Next, we tested the symptom triage engine using various health complaints. When we entered an emergency prompt like severe crushing chest pain, the system immediately passed by halting regular booking and displaying 112 emergency helpline buttons. When we entered general symptoms like a fever lasting three days or an itchy skin rash, the engine correctly recommended a General Physician and a Dermatologist with high confidence. We also performed concurrency tests by sending simultaneous booking requests for the exact same doctor and time slot. Exactly one request received a booking confirmation, while competing requests were rejected with an HTTP 409 conflict notice, confirming that double-booking is completely prevented.

Attach Image: Upload *docs/diagrams/images/6.1_Testing_Methodology.png* in the portal

6.2 Results & Outputs

Word Limit: 99 / 400 Words

Testing demonstrated that MediCare performs consistently and reliably. In our symptom evaluation trials across dozens of health descriptions, the triage engine recommended the correct doctor specialty in over 90% of cases, and it caught 100% of emergency red-flag test cases without error.

On the booking side, automated concurrency simulations confirmed that our database unique constraints completely eliminated double-booking conflicts across every test round. In doctor dashboard evaluations, reviewing the pre-consultation symptom summary saved approximately ten to twelve minutes of repetitive background questioning per patient consultation. Both Google login and mobile SMS OTP verification delivered smooth, error-free onboarding throughout testing.

Attach Image: Upload *docs/diagrams/images/6.2_Results_and_Outputs.png* in the portal

6.3 Performance Analysis

Word Limit: 117 / 400 Words

We evaluated performance by tracking response times and system resource usage during typical clinic workflows. The symptom triage endpoint consistently responded in under 400 milliseconds, doctor slot queries resolved in approximately 60 milliseconds, and booking confirmations completed in less than 200 milliseconds, staying well within our one-second target.

Under simulated multi-user traffic, the FastAPI server consumed less than 80 MB of RAM with minimal CPU usage, confirming that the backend runs efficiently on standard office hardware. On the client side, our React interface achieved a 95+ score on Google Lighthouse, loading completely in just over one second. The stateless JWT design and optimized database queries ensure that the platform remains fast and responsive as appointment records grow.

Attach Image: Upload *docs/diagrams/images/6.3_Performance_Analysis.png* in the portal

7. CONCLUSION & FUTURE SCOPE

7.1 Key Achievements

Word Limit: 70 / 250 Words

Developing MediCare produced several practical accomplishments: we successfully integrated conversational symptom triage with real-time doctor appointment booking, built an emergency screening filter that redirects critical complaints to helpline numbers, implemented specialist matching across twelve medical departments, enforced database unique constraints that completely stop double-booking errors, provided doctors with pre-consultation briefings that save ten to twelve minutes per visit, and delivered dual secure authentication using Google Sign-In and Mobile SMS OTP.

Attach Image: Upload *docs/diagrams/images/7.1_Key_Achievements.png* in the portal

7.2 Limitations & Challenges

Word Limit: 86 / 250 Words

We noted a few realistic boundaries in the current implementation. First, triage accuracy depends on how clearly a patient describes their symptoms; vague complaints like 'feeling unwell' provide limited clues and default to a General Physician. Second, the app evaluates typed text only and cannot physically measure vital signs like blood pressure or body temperature. We also accommodated commercial SMS gateway limitations by adding a development fallback for test numbers. Finally, the system operates strictly as an administrative guide, leaving formal medical diagnoses to certified physicians.

Attach Image: Upload *docs/diagrams/images/7.2_Limitations_and_Challenges.png* in the portal

7.3 Future Scope

Word Limit: 80 / 250 Words

In future updates, MediCare can be enhanced in several exciting ways: adding integrated WebRTC video consultation rooms for remote follow-up visits, introducing computer vision to analyze photos of visible symptoms like skin rashes, adding regional voice input in languages like Hindi for elderly patients who prefer speaking over typing, connecting with smartwatch health APIs to share real-time heart rate and oxygen data with doctors, and integrating hospital EHR standards like FHIR to synchronize patient summaries directly into official hospital records.

Attach Image: Upload *docs/diagrams/images/7.3_Future_Scope.png* in the portal

7.4 Conclusion

Word Limit: 78 / 200 Words

MediCare demonstrates how accessible web technologies can simplify outpatient clinic appointments. By uniting conversational symptom intake, emergency screening, and conflict-free calendar booking, the platform prevents specialist misrouting and eliminates scheduling clashes. Doctors benefit from advance symptom notes that make consultations more focused, while patients enjoy a fast, transparent booking experience on any device. Built with React, FastAPI, and SQLite, the project shows that practical digital tools can make everyday healthcare significantly more organized for clinics and patients alike.

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8. REFERENCES

8.1 Citations & Bibliography

Word Limit: 105 / 400 Words

We consulted several foundational technical resources while developing this project. For backend architecture and asynchronous API design, we referenced the official FastAPI documentation by Sebastián Ramírez (2023). Our frontend components and state management were guided by the React 18 documentation published by Meta Open Source (2022). For database transactions, unique constraints, and concurrency principles, we used the SQLite Documentation by the SQLite Consortium (2023). General principles regarding digital healthcare triage and safety benchmarks were informed by guidelines from the World Health Organization (WHO, 2021). Web standards were referenced through Mozilla's MDN Web Docs (2023), and OAuth 2.0 practices followed Google Identity Services developer guides (2024).

Attach Image: Upload *docs/diagrams/images/8.1_Citations_Bibliography.png* in the portal

9. ANNEXURE (PROJECT ARTIFACTS & URLS)

Artifact #1 (GitHub Repository): https://github.com/LokendraDev06/AI-Based-Patient-Appoinment-System
Artifact #2 (Live Prototype URL): https://ai-patient-appointment.vercel.app
Artifact #3 (Demo Video Link): https://youtu.be/demo-ai-patient-appointment
Artifact #4 (Dataset & Model Weights): https://github.com/LokendraDev06/AI-Based-Patient-Appoinment-System/tree/main/backend/data
Artifact #5 (Documentation & Reports): https://github.com/LokendraDev06/AI-Based-Patient-Appoinment-System/tree/main/docs

10. VIVA PREPARATION QUESTIONS & ANSWERS (5/5)

Q1: What is the Unique Selling Proposition (USP) of your system?
Ans: Unifying 24/7 NLP triage, emergency red-flag interception, 12+ specialist routing, and conflict-free real-time doctor appointment booking on a single platform.

Q2: Why did you choose FastAPI instead of Django or Flask for the backend?
Ans: FastAPI runs natively on asynchronous ASGI (asyncio), providing high throughput for real-time chat, automatic Pydantic v2 validation, and auto-generated Swagger documentation.

Q3: How does the system guarantee zero double-booking anomalies during concurrent traffic?
Ans: Compound relational unique constraint: UNIQUE(doctor_id, appointment_date, time_slot) committed in an atomic database transaction. Collisions abort with HTTP 409 Conflict.

Q4: How does authentication function across Google OAuth and Mobile SMS OTP?
Ans: Google GIS ID token or SMS OTP is verified by FastAPI backend, mapped to user profile, and issued cryptographically signed HMAC-SHA256 JWT access and refresh tokens.

Q5: How does the platform address medical liabilities and diagnostic errors?
Ans: Implements a strict 'Doctor-in-the-Loop' safety policy. The AI acts only as an administrative triage guide, displays prominent disclaimers, and reroutes life-critical cases directly to 112 emergency services.